The Theorem Ladder of Resolution Geometry (v2)

A Consolidated Structural Leg: What Is Forced, Declared, Imported, and Open,

from the Kernel-Pair Root to the Observable Cone

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Abstract

The corpus has grown into many separate hostile-attack papers—kernel pair, quotient, assembly, reconstruction, ignorance, observation limit, saturation, cone, scale, transport, duration. This note supplies the missing independent leg: a single dependency-ordered ladder that states, for the whole framework, exactly what forces what. It now adds one theorem-level typing correction forced by a dedicated hostile battery: the observable cone cannot jump directly to Lorentzian conformal reconstruction. A projective realization module must first separate disguised quadratic, Lorentz–Finsler, multicone, stratified, and residual characteristic branches. Version 2 adds Stratum 7, the final remainder: the information residual and its terminal refinement, the coupling stabilizer, the stack of constant-type families, the Global Structure Theorem identifying the remainder with local canonical-correlation moduli plus a tuple of Euclidean vector bundles, its universality, the $\check{H}^2$ gluing level, and the wall boundary; it corrects R9 and R21 and records two refuted verdicts. The remaining content is the *typing*: every rung is marked **forced** (by the exact kernel pair), **imported** (established mathematics reused), **gated** (valid only under an explicit hypothesis), **declared** (added structure, not forced), **no-go** (a forced negative), or **open** (a conjecture/gap). The spine is: an observation map F forces its kernel pair R_F, and R_F forces—and forces *only*—the exact extensional layer: observational equivalence, the quotient $Q_F \cong \operatorname{Im} F$, fibre saturation, and universal factorization of fibre-constant constructions. Every higher structure crosses a type boundary and needs its own gate: the tangent shadow $\ker dF$ (regularity), the second-order covariance/Fisher shadow (a declared metric; Fisher provably insufficient), the statistical order (Blackwell $\neq$ kernel-pair order), assembly completeness (a sound groupoid need not generate every fibre), the observable cone (a declared fibre-compatible reachability plus a topology and a Lorentzian-realization gate, after which HKMM reconstruction gives a conformal class), scale (an integrable local clock field; a single worldline underdetermines it), and transport (the induced connection on regular strata, with the canonical connection the largest open gap). The kernel-pair gap is therefore mapped precisely: the root is universal for exact information and for nothing else, and each extension is a named module with a named hypothesis. The result is a leg one can audit rung by rung, with the frontier—connection reconstruction, Blackwell realizability, singular strata, and the physical scale anchor—marked as such.

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1 Why a leg, and how to read it

Each existing paper attacks one joint. What a foundation additionally needs is a spine: the ordered list of load-bearing steps, each tagged by *what kind* of statement it is, so a reader can see where mathematics forces the structure, where a hypothesis is silently doing the work, and where the frontier lies. The tags are:

forced	compelled by the exact kernel pair R_F alone
imported	established mathematics reused, not RG-original
gated	valid only under an explicitly stated hypothesis
declared	added structure, chosen not forced
no-go	a forced negative result (something cannot be done)
open	conjecture or gap; not established

The organizing fact, established by the kernel-pair-root attack, is that only the first stratum below is *forced*; every later rung crosses a type boundary.

2 The ladder

Rung	Statement	Type	Depends on
Stratum 0 — the exact root (the only forced layer)			
R0	Observation map $F : X \to Y$ declared	input	—
R1	Kernel pair $R_F = X \times_Y X$ is an equivalence relation (exact ignorance)	forced	R0
R2	Quotient $Q_F = X/R_F \cong \operatorname{Im} F$; every fibre-constant map factors uniquely through Q_F	forced	R1
R3	Saturation: observable invariants are exactly R_F-invariant functions on Q_F	forced	R2
Stratum 1 — assembly (bottom-up = top-down, conditionally)			
R4	Kernel groupoid $K_{\text{obs}} = X \times_{Q_F} X$; R_F is a groupoid	forced	R1
R5	Declared automorphism groupoid A; sound $A \hookrightarrow K_{\text{obs}}$ gives surjection $X/A \to Q_F$	gated (soundness)	R4
R6	$X/A \cong Q_F$ iff A is gauge-complete (generates every fibre)	gated (completeness)	R5
Stratum 2 — second-order shadow (metric detection)			

Rung	Statement	Type	Depends on
R7	Regular reconstruction: (R, N, Σ) is a canonical gauge-equivariant function of (J, Σ, g)	imported	R2
R8	Fisher $= J^\top \Sigma^{-1} J$ = inverse Schur = second-order tangent shadow	imported	R7
R9	Fisher insufficiency: blind to the null block at $C_{RN} = 0$ *relative to a declared complement* N (C and $L = CD^{-1}$ move affinely under a change of complement, G and D do not); does not reconstruct the covariance; globally the Fisher-relative remainder is exactly $[K_N] \in \mathrm{Vect}_{q-r}(S)$ (R38)	no-go (gated: H-split)	R8, R35
R10	Tangent caveat: $\ker dF$ = tangent of the fibre only on regular strata; higher contact is jets of F, not covariance	gated (regularity)	R2,R8
Stratum 3 — ignorance and the observation limit			
R11	On regular models $\mathrm{rad}\, g^F = \ker dF$ (Fisher detects, does not define, ignorance)	gated (regularity)	R8,R10
R12	Persistent null $\mathcal{I}_x = \ker dF_\infty$ over a compatible sensor family; attained finitely (finite dim)	gated (compatibility)	R11
R13	Three layers: sensor = wall $\mathcal{I}_x$; precision (evaluation) and representation (downstream) = removable	forced (typing)	R12
R14	Kernel-pair order $\neq$ Blackwell order; equal identifiability kernel pairs need not be Blackwell-equivalent	no-go	R1
Stratum 4 — declared structure on the quotient			
R15	Fibre-compatible reachability $\preceq$ descends to a preorder on Q_F; antisymmetrize to a causal order	declared	R2
R16	Future sets J^+_{obs}; a cone boundary requires a supplied compatible topology	gated (topology)	R15
R17	Projective-cone invariance: classify the zero set/direction boundary, not the degree or chosen defining function x	no-go / typing	R16
R18	Quadratic-cone gate: equality with the null cone of a nondegenerate Lorentzian quadratic form gives a unique positive conformal ray $[g_{\mathrm{obs}}]$	imported (gated)	R17
R19	Lorentz–Finsler gate: a single smooth, time-oriented, strongly convex homogeneous cone with nondegenerate Lorentzian fibre Hessian admits a direction-dependent realization	gated	R17
R20	Multicone gate: persistent multiple projective characteristic sheets require a multicone or multi-Finsler realization; one conformal metric is insufficient	gated	R17

Rung	Statement	Type	Depends on
R21	Stratified-realization gate: rank, isotropy, signature, topology, or sheet-count changes require regular strata joined by slices/tangent cones; failures of homogeneity or convexity remain general characteristic/admissibility geometry. A transversal codimension-2 collision of two simple canonical correlations *forces* $w_1 \neq 0$ of the colliding singular line bundles around the wall (R41)	gated / open globally	R17–R20, R41
R22	Scale: $\mathcal{T} = -\log C$ additive on the zero-memory stratum; a single worldline underdetermines the conformal factor	gated	R17
R23	Metric representative exists iff local scale data glue as an exact cocycle (zero scale-holonomy)	gated (integrability)	R22
R24	Cross-stratum (astronomy): remote null test after a correspondence map and nuisance model; ratio-only identifiability	gated	R17,R23
Stratum 5 — transport, and the main open gap			
R25	On a regular metric stratum with projector P: $\nabla^W = P\,\nabla^{\mathrm{LC}}$; second form $\mathrm{II} = (I-P)\nabla^{\mathrm{LC}}$	imported (gated)	R7,R17
R26	Entropy no-go: $H(X), H(Q), H(X \mid Q)$ are fibre-automorphism invariant; entropy cannot select a connection	no-go	R2
R27	Canonical connection from (J, Σ, K) (memory kernel required): $(J, \Sigma, K) \to [\nabla]$	open	R25,R26
Stratum 6 — duration algebra (self-contained)			
R28	$\mathcal{T} = -\log C$; memory = BCH defect $\frac{1}{2}[\mathcal{T}, \mathcal{T}] + \cdots$; additive iff commuting	imported	R22
R29	Scalar clocks = trace character $\tau = a\,\mathrm{tr}\,\mathcal{T}$, unique up to unit (Cauchy)	imported	R28
Stratum 7 — the final remainder (information, symmetry, descent, walls)			
R30	Task residual $\mathcal{R}_y = \{a(x) : c(x) = y\}$; tasks descend through c iff every $\mathcal{R}_y$ is a singleton	forced (contract)	R0–R2
R31	Terminal sufficient refinement $E = \mathrm{im}(c, a)$: unique up to unique isomorphism, natural in contract-preserving maps	forced	R30
R32	Joint witnesses required: marginal possibility sets lose correlation	no-go	R30
R33	Operational closure = $\ker(\beta : X \to Z^M)$, the greatest M-stable congruence (Myhill–Nerode, bisimulation)	imported	R31
R34	Smooth descent of fibre-constant maps iff c is a submersion; x^3 counterexample otherwise	gated (regularity)	R31

Rung	Statement	Type	Depends on
R35	Coupling stabilizer $\mathrm{Aut} \cong \prod_j \mathrm{O}(m_j) \times \mathrm{O}(p-r) \times \mathrm{O}(q-r)$; $\dim \mathrm{Aut} = \sum \binom{m_j}{2} + \binom{p-r}{2} + \binom{q-r}{2}$, $\pi_0 = (\mathbb{Z}/2)^\kappa$; multiplicities, not rank, control symmetry	imported (SVD)	R7
R36	RG families of constant multiplicity type form a stack	forced (H-mor)	R35
R37	Twisted forms $\leftrightarrow \check{H}^1(S; \mathrm{Aut}_x)$; a pointed set whose base point is the chosen reference	imported (Giraud, Serre)	R36
R38	Global Structure Theorem: $\mathrm{RG}_\mu(S) \simeq C^\infty(S, \Delta_k) \times \prod_j \mathrm{Euc}_{m_j}(S) \times \mathrm{Euc}_{p-r}(S) \times \mathrm{Euc}_{q-r}(S)$, natural in S	proved (gated: H-type)	R35–R37
R39	Universality: $(\sigma, [E_\bullet], [K_R], [K_N])$ is the initial complete natural invariant; homotopy part represented by $B\,\mathrm{Aut}_\mu$, unique up to homotopy equivalence	proved	R38
R40	Second level: triple defects with an abelian band define a class in $\check{H}^2$; removable iff zero (filled triangle removable, $\mathbb{R}P^2$ not)	imported (gated: abelian band)	R36
R41	Isotropy jump: no single-model (principal-bundle) description on any neighbourhood of a type change; forced wall monodromy at transversal simple collisions	no-go / proved (gated)	R35
R42	Classification across walls (exit paths, constructible stacks)	open	R38, R41
R43	No primitive: every component of the remainder is a named classical construction with a universal property	proved (gated: H-split, H-type, H-mor)	R30–R40
R44	"No faithful functor into any single foundation" and "intrinsically nonzero descent class"	refuted	R38

3 The kernel-pair gap, stated exactly

The single most important structural fact is where the root stops.

Gap 1 (What R_F forces, and what it does not). The exact kernel pair forces exactly R1–R4: observational equivalence, the quotient, fibre saturation, universal factorization, and the kernel groupoid. It forces *nothing else*. Each of the following is a separate type and requires its own hypothesis:

(i) the *tangent* shadow $\ker dF$ equals the fibre tangent only on regular strata (R10); at singular points it does not.

(ii) *higher contact* is governed by jets of F, not by any covariance; formal flatness need not imply exact local constancy (R10).

(iii) *Fisher/covariance* is a declared distinguishability structure, not kernel-pair data, and is insufficient to reconstruct the covariance (R8–R9).

(iv) the *statistical order* (Blackwell/deficiency) is strictly finer than sameness of identifiability kernel pair (R14).

(v) *assembly* closes only under gauge completeness; a sound groupoid need not generate every fibre (R6).

(vi) *precision* and *representation* act on evaluation and on downstream constructions, not on R_F (R13).

These six crossings are the kernel-pair gap. Naming them is the leg's central service: it forbids smuggling any higher module into the root.

Gap 2 (The main open rung). R27—a canonical connection from (J, Σ, K)—is the largest remaining source of non-uniqueness in the characterized object. The entropy no-go (R26) shows a natural candidate selector fails; the memory kernel (R28) is the extra datum the reconstruction needs; the theorem is not yet in hand. Everything downstream of a connection (curvature, holonomy, stratified transport) inherits this gap.

Gap 3 (The final remainder, stated exactly). On strata of constant multiplicity type the remainder is locked (R38–R39): it is the pair (local moduli σ, global class of the block bundles), the initial complete natural invariant, built entirely from classical constructions (R43). It is *not* a primitive and *not* an intrinsic property of RG: over a contractible base the global class vanishes; over S^1 it is $(\mathbb{Z}/2)^\kappa$. Two corpus verdicts are withdrawn (R44); the rigidity proposition of the Lock Conclusion is replaced by R35 (witness $C = \frac{1}{2}I_3$: rank 3, $\dim \mathrm{Aut} = 3$). The unlocked part is R42: classification across walls, where R41 shows that monodromy is forced.

Gap 4 (The physical anchor is outside the ladder). Absolute scale (R22–R23) and the identification $v_\partial = c$ are declared/empirical, not forced by any rung. This is why the cone yields $[g]$ but not g: the ladder is conformal-complete and scale-open by construction. The anchor is a physical input, correctly placed outside the forced spine.

4 The minimal "must" path to the observable cone

Extracting only the load-bearing rungs, the shortest sound route from an experiment to an observable characteristic geometry is

$$\underbrace{F}_{\text{R0}} \to \underbrace{R_F \to Q_F}_{\text{R1–R2, forced}} \to \underbrace{\preceq_{\text{obs}}}_{\text{R15, declared}} \to \underbrace{\text{topology and cone boundary}}_{\text{R16, gate}} \to \underbrace{\text{projective realization test}}_{\text{R17}}.$$

The realization test branches rather than continuing automatically:

$$\begin{aligned}
\text{quadratic regular cone} &\xrightarrow{\text{R18}} [g_{\text{obs}}],\\
\text{single smooth strongly convex cone} &\xrightarrow{\text{R19}} \text{Lorentz–Finsler target},\\
\text{persistent multiple sheets} &\xrightarrow{\text{R20}} \text{multicone target},\\
\text{type-changing discriminant} &\xrightarrow{\text{R21}} \text{stratified tangent-cone/slice target}.
\end{aligned}$$

If homogeneity, regularity, or convexity fails, R21 retains only general characteristic or admissibility geometry. Groupoid assembly (R4–R6) remains an alternative certificate for R2. The second-order stratum (R7–R11) is required only for metric/ignorance geometry. Scale (R22–R24) and transport (R25–R27) are further modules. The quadratic path is therefore not universal; it is one gated branch of a larger realization rung.

5 Final cone-realization attack and reproducibility

The deterministic artifact `rg_cone_realization_final_attack.py`, seed `20260925`, registered and passed all 13/13 hostile checks. The machine-readable record is `rg_cone_realization_final_attack.json`. The attacks were:

1. a degree-six presentation $x = Q^3$ with exactly the same projective null cone as a Lorentzian quadratic form, rejecting polynomial degree as a realization invariant;
2. a smooth homogeneous single-sheet witness that rejects every fitted quadratic cone on held-out directions while retaining a positive spatial fibre Hessian and Lorentzian fundamental-tensor signature, forcing the Lorentz–Finsler branch;
3. a product of two separated Lorentzian factors, producing two persistent characteristic sheets that cannot be represented by one nondegenerate Lorentzian quadratic form, forcing the multicone branch;
4. a parameterized sheet collision whose sheet count changes $2 \to 1 \to 2$ and whose characteristic gradient vanishes at the collision wall, forcing a stratified transition locus;
5. nonhomogeneous and homogeneous-but-nonconvex residual witnesses, proving that the named branches are gated and are not an exhaustive theorem without further hypotheses.

The analytic conclusion is stronger than a numerical fit: the explicit witnesses are counterexamples to the previous universal jump from R16 to Lorentzian conformal reconstruction. Hence R17–R21 are logically necessary typing rungs. The battery does *not* prove that Finsler, multicone, and stratified geometry exhaust every possible characteristic object; the residual branch is retained explicitly.

6 Final-remainder attack and reproducibility

The deterministic artifact `rg_final_remainder_attacks.py`, master seed `20261022`, runs 18 attacks and 4 canaries (all canaries rejected; two fresh-process runs give identical witnesses). Status: 5 refuted, 6 sharpened, 5 boundary, 2 confirmed. The machine-readable record is `rg_final_results.json`; the theory is in *The Final Remainder of Resolution Geometry*. In the final print *Resolution Geometry*, R30–R44 correspond to rungs L116–L123 and the open rung L126. Load-bearing witnesses for Stratum 7: brute-force gauge orbits (S_3: $1, 3, 11$ classes on tree, circle, figure eight), exact rational stabilizer dimensions on 22 non-whitened instances, the Möbius RG family with constant local invariants, all 2^{15} lifts on $\mathbb{R}P^2$, the componentwise Čech computation on a non-Leray cover, and the wall-monodromy sign.

7 Non-claims

Non-claim 1 (No exhaustive classification theorem). The new realization module proves that an intermediate rung is necessary and supplies explicit witnesses for its principal branches. It does not claim that Lorentz–Finsler, multicone, and stratified cones exhaust all homogeneous characteristic geometry. The residual branch in R21 is intentional.

Non-claim 2 (No new primitive). The root is a kernel pair; the imported rungs are Schur/Fisher, HKMM reconstruction, BCH, Cauchy's equation, SVD, descent and classifying spaces. On constant-type strata this is now a theorem (R43) under H-split, H-type and H-mor, not only a remark; across walls (R42) it is open, and open is not primitive.

Non-claim 3 (Conformal, not metric; observational, not dynamical). The forced and gated rungs reach a conformal class and an observational transport at most. Absolute scale, the light-speed identification, and Einstein dynamics are outside the ladder.

Non-claim 4 (Gaps are gaps). R14 (Blackwell), R21 (global stratified compatibility), R27 (canonical connection), R42 (classification across walls), and the physical anchor are open or external and are not asserted as closed.

8 Conclusion

The corpus has an independent leg: a spine of forty-five rungs in eight strata, each typed, with the exact kernel pair forcing only the first four and every later structure carrying a named hypothesis. The observable cone is reached by a short forced-plus-gated path; the metric, the transport, and the physics are further modules, each gated or open. The two structural facts worth carrying are that the kernel pair is universal for exact information *and nothing else*—the six type crossings are the gap—and that the single largest remaining non-uniqueness is the canonical connection R27. A framework that can state its own dependency graph, its imports, and its open rungs this precisely is one whose claims can be audited; that auditability, not any new theorem, is what this leg supplies.

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